Duke University
Dr. Christopher Woods
STATEMENT OF WORK
Title – PREdicting contagion using Systems And GEnomic analysis (PRESAGE)
Date: 11 November 2016
Updated: 14 June 2018 (P00004)

1.0 SCOPE

The scope of this project, as part of the Prometheus program, is to advance the understanding of host molecular features that are predictive of infectious disease spreading potential by performing human infectious disease studies; generating viral, socio-environmental, and molecular-based data from said studies; setting up informational systems to store, process, analyze, and transmit data; and performing analytics on retrospective and prospective samples or datasets to discover the molecular and environmental nature of host transmission potential and symptomatic response to pathogen with the aim of identifying a minimal set of early host biomarkers in humans that correlate with and predict contagiousness within 24 hours after infection.

1.1 BACKGROUND

The Prometheus program is described in DARPA-BAA-16-42 as follows:

"The Prometheus program aims to discover molecular host prognostic biomarkers that predict if a person will become contagious after exposure to a respiratory pathogen. The spread of many acute infections is caused by people who are contagious prior to developing symptoms. By the time an individual develops symptoms and seeks medical attention, they have likely spread the disease to close contacts. Furthermore, since these contagious individuals may not seek health care (because their symptoms are mild), they are not counted in disease surveillance activities. This fundamental lack of knowledge of disease transmission in a community hinders our ability to characterize an outbreak and forecast what will happen next. Relying on clinical diagnostic information will typically result in late and inaccurate models of disease transmission, making it impossible to forecast the course of an outbreak. Identifying contagious individuals early is a critical capability for outbreak mitigation."

With the continuing advancement of molecular base tools has come the ability to generate tremendous quantities of data that can be used to identify and quantify biological markers indicative of exposure, disease, and contagiousness. The ability to collect and analyze this data presents the possibility of predicting the contagiousness of pre-symptomatic individuals through the monitoring of key molecular signatures.
This program aims to discover host molecular features that predict infectious disease transmission potential. The end goal of the program is a minimal set of early human host biomarkers that predict contagiousness within 24 hours of infection or exposure to pathogen.

Towards accomplishing the Prometheus program's aims, this effort being proposed by a team consisting of Duke University (DU), Imperial College London (ICL), University of Michigan (UM), and University of North Carolina at Chapel Hill (UNC) will focus on gathering and analyzing samples and data from retrospective and prospective influenza challenge studies. The effort proposes to 1) conduct a human influenza challenge study to collect samples and data at relevant time points for assessing contagiousness in pre-symptomatic subjects, 2) analyze retrospective and prospective samples to generate biomarker data at these relevant time points, 3) establish a database for data generated as part of the study, and 4) conduct mathematical analysis of the biomarker data to generate predictors of contagiousness. The measurement of success is to achieve an accuracy of 95% in predicting contagiousness within one day of exposure to a pathogen.

2.0 APPLICABLE DOCUMENTS

2.1 DARPA BAA-16-42.
2.2 Duke University Technical Proposal Titled Predicting Contagiousness of Influenza Infection Prior to Symptom Display dated 11/11/2016.

3.0 Project Work Description and Requirements

The recipient shall provide the facilities and equipment necessary to develop the effort as described herein.

Human subjects research (HSR) is anticipated as part of this effort. Prior to release of funds associated with HSR, the recipient shall obtain all necessary Institutional Review Board (IRB) approvals, show proof of proper assurance documentation (Federal-Wide Assurance (FWA)), and obtain proper approval from the Government officials (SSC HRPO) prior to initiation of human subject testing and release of funds. All amendments and/or renewals as well as reports of non-compliance, Unanticipated Problems that May Involve Risks to Subjects or Others (UPIRTSO), Adverse Event (AE), and/or audits conducted to any protocol under this effort will also require approval by government officials (SSC-HRPO) prior to proceeding under said changes. A final report closing out ALL protocols will also be required either at close of HSR related work and/or prior to end of agreement whichever comes first. For assistance with preparing IRB documentation for review and approval by SSC-HRPO contact the SSC program manager.

Animal use is NOT anticipated in this effort. However, if this changes a modification to the cooperative agreement will be required and the recipient shall obtain all necessary Institutional Animal Care and Utilization Committee (IACUC) approval, along with secondary approval by a DOD IACUC and demonstrate this approval to the Government (SSC-Pacific program manager) prior to beginning experimentation with animals. The recipient shall also obtain approval from both Animal Care and Use Review Office (ACURO) and SSC-Pacific on all amendments and/or renewals prior proceeding under said changes. For assistance with submission of animal research related documents, contact the program's manager.

3.1 BASE PERIOD (Phase I)

3.1.1 DU - Generate pilot metabolomics data.

TASK 3.1.1 SHALL NOT BE EXERCISED AND TASKING FUNDS RELEASED UNTIL IRB DOCUMENTATION AND PROPER IRB APPROVAL HAS BEEN OBTAINED.

This task consists of conducting pilot metabolomics analyses from existing Duke (and other performers) cohorts of Influenza-challenged samples. The pilot analysis results are intended to inform the down-selection of metabolomics assays and time points as further described in Task 3.1.6.

3.1.1.1 The recipient shall identify and compile existing samples at relevant time points from Duke cohorts.

3.1.1.1.1 (3 Month) Milestone: Curated list of existing samples from Duke cohorts.

3.1.1.2 The recipient shall evaluate quality standards of samples from the DARPA Prometheus program consortium and availability of relevant phenotypic and molecular information including host response signal(s) for potential inclusion in the pilot analyses.

3.1.1.2.1 (3 Month) Milestone: Curated list of existing samples from possible consortium cohorts.

3.1.1.3 The recipient shall generate data on 3 or more metabolomics platforms (may include but not limited to Absolute IDQ p180 (Biocrates), Unbiased lipidomics, targeted Oxylipin/Oxysterols, energy metabolite panel, and/or purine/pyrimidine panel) on existing flu samples (number of samples = 100; 25 samples at 4 time points). The recipient shall consult Government on choice of metabolomics platforms.

3.1.1.3.1 (3 Month) Milestone: Pilot data generated on 3 or more metabolomics platforms (may include but not limited to p180, lipids, oxylipin/oxysterols, energy metabolite panel, and/or purine/pyrimidine panel.)

3.1.1.4 The recipient shall conduct quality control (QC), normalize metabolomics data, and package generated datasets for analysis as described in Task 3.1.6.

3.1.1.4.1 (3 Month) Milestone: QC, normalized, and packaged pilot dataset for analyses.

3.1.1.4.2 (3 Month) Deliverable: Fully curated and annotated pilot dataset generated and ready for analyses in Task 3.1.6.

3.1.2 DU – Design and build federated database structure of Duke data.

The recipient shall design and implement a database architecture leveraging existing infrastructures to host all data (phenotypic and molecular) generated by the Duke team as part of the Prometheus program. In addition to leveraging current standard operating procedures (SOPs) and curation pipelines, the recipient shall develop SOPs and customize pipelines for data handling, data dictionaries and sharing protocols for all datasets (retrospective and prospective) specific to supporting the DARPA Prometheus program.

3.1.2.1 The recipient shall design SOPs for data handling, database design, and structure and content of data dictionaries

3.1.2.1.1 (4 Month) Milestone: Developed SOPs for data handling, database design, structure, and content of dictionaries.

3.1.2.2 The recipient shall implement and test database.

3.1.2.2.1 The recipient shall perform standard tests to ensure access, data integrity, and compliance with design and data dictionary specifications.

3.1.2.2.2 (4 Month) Milestone: Database implementation and testing complete.

3.1.2.3 The recipient shall develop sharing protocols, launch database, and implement user protocols.

3.1.2.3.1 (4 Month) Milestone: Sharing of user protocols developed and launched.

3.1.2.4 The recipient shall populate database with retrospective data from Task 3.1.1 and shall perform standard checks to ensure data integrity.

3.1.2.4.1 (4 Month) Milestone: Database populated and tested with pilot dataset from Task 3.1.1

3.1.2.4.2 (4 Month) Deliverable: Functional and access-secured federated database generated.

3.1.3 DU, ICL, UNC – Execute Influenza challenge study (number of subjects = 40)

TASKS 3.1.3.3 – 3.1.3.4 SHALL NOT BE EXERCISED AND TASKING FUNDS RELEASED UNTIL IRB DOCUMENTATION AND PROPER IRB APPROVAL HAS BEEN OBTAINED.

The Influenza challenge study shall be performed under a subcontract collaboration with Imperial College, London. To assure appropriate regulatory compliance, required approval of the study protocol and supporting documents shall be obtained from the local UK ethics committee, Duke IRB and HRPO review boards prior to subject enrollment.

3.1.3.1 IRB Documentation Generation: The recipient shall obtain all necessary IRB documentation and provide to Government in accordance with IRB documentation submission guidance prior to conducting human studies.

3.1.3.2 The recipient shall obtain regulatory approvals from local UK ethics committee, Duke IRB board, and HRPO.

3.1.3.2.1 (8 Month) Milestone: Regulatory approvals obtained from UK Ethics, Duke IRB, and HRPO

3.1.3.3 The recipient shall screen up to 250 individuals to assess pre-existing immunity to pH1N1(2009) virus by hemagglutination assay (HA).

3.1.3.3.1 (8 Month) Milestone: Up to 250 subjects screened for pH1N1 antibodies by HA.

3.1.3.4 The recipient shall challenge up to 40 young, healthy volunteers inoculated with pH1N1 (2009) at Imperial College London. (Total Quarantine Time: 12 weeks. Attack rate estimated at 50% with resulting VL 2-4log10 copies/mL)

3.1.3.4.1 The recipient shall sample whole in blood in paxgene tubes for RNA stabilization and EDTA tubes for peripheral blood mononuclear cell (PMBC) isolation at 6 hour intervals between 12 and 48 hours after inoculation with subsequent sampling occurring every 12 hours through Day 5 after inoculation, or if necessary, the recipient may adjust sampling times and intervals to coordinate with other DARPA Prometheus performers. The recipient shall determine final collection schedule by sample type, meaningful timepoints, clinical collection constraints, and cost limits. The recipient shall collect convalescent blood samples at Day 28 after inoculation.

3.1.3.4.2 The recipient shall collect nasal wash/swab sample at 12 hour intervals beginning at 24 hours after inoculation through Day 5 after inoculation and will collect an additional sample at Day 28 after inoculation. If necessary, the recipient may adjust sampling times and intervals to coordinate with other DARPA Prometheus performers.

3.1.3.4.3 The recipient shall collect nasal cell samples, representing the most proximal viral:host interaction site using curettage at baseline, 24, and 48 hours after inoculation; and at convalescence at Day 28 after inoculation. If necessary, the recipient may adjust sampling times and intervals to coordinate with other DARPA Prometheus performers.

3.1.3.4.4 The recipient shall collect physiologic monitoring (empatica) and auditory monitoring (Opo) data to augment biomarker and clinical symptom data to enhance performance of predictive models.

3.1.3.4.5 (8 Month) Milestone: Up to 40 subjects challenged with pH1N1, and time-series bio-specimens, clinical, physiologic, audio, and laboratory data collected.

3.1.3.4.6 (8 Month) Deliverable: Study samples and associated clinical, physiologic, and laboratory data secured and housed in Duke biobank and data system.

3.1.4 DU – Generate molecular data from prospective sample collections at meaningful time points.

THE TASK (3.1.4) SHALL NOT BE EXERCISED AND TASKING FUNDS RELEASED UNTIL IRB DOCUMENTATION AND PROPER IRB APPROVAL HAS BEEN OBTAINED.

Generate molecular (biomarker) data. The recipient shall use an adaptive strategy in selecting samples for assaying and analysis; the number of subject samples and time points to analyze will depend on the clinical information and informative time points needed to achieve the program objectives. Previously, the recipient extensively demonstrated that blood-based biomarkers are capable of accurately diagnosing acute respiratory viral infections. However, to date, the technologies for detecting these biomarkers have demonstrated low signal-to-noise ratios at 24-hours post-exposure. Therefore, in order to increase sensitivity to identify these individuals with 95% accuracy at this critical time point the recipient shall use combination of existing and novel technologies to accomplish this.

3.1.4.1 The recipient shall prepare samples for RNA sequencing (RNASeq)

3.1.4.1.1 The recipient shall perform RNA extraction of peripheral whole blood (up to 260 samples;) for transfer to collaborators for RNASeq data generation.

3.1.4.1.2 The recipient shall isolate and stabilize RNA from single immune cell preparations from peripheral blood ~~for transfer~~ and shall perform RNA sequencing of the preparations at ~~to collaborators~~ the Duke Molecular Physiology Institute ~~for single cell RNASeq~~ (up to 60 samples; a goal of 20 subjects at 3 timepoints).

3.1.4.1.3 The recipient shall isolate RNA from nasal curettage for RNASeq (up to 200 samples)

3.1.4.1.4 The recipient shall perform RNASeq data generation from total RNA isolated from nasal curettage samples (up to 120 samples) using Illumina RNASeq shall be conducted at the Duke Sequencing Core Lab at appropriate specifications, specifically total RNA (mRNA and non-coding RNAs) at 75M reads per sample.

3.1.4.1.5 (12 Month) Milestone: RNA extractions prepared from peripheral blood, PBMCs, ~~and~~ nasal curettage samples for RNASeq~~.~~, and complete RNA sequencing from single immune cell preparations from peripheral blood.

3.1.4.2 The recipient shall analyze cytokine expression

3.1.4.2.1 The recipient shall conduct a 30-analyte Mesoscale Discovery (MSD) panel on peripheral blood and lavage samples generated from the challenge (200 samples; 40 subjects *at 5 time points)*

3.1.4.2.2 *(12 Month) Milestone: 30-panel MSD assays completed on peripheral blood and nasal lavage samples.*

3.1.4.3 The recipient shall conduct an enriched phosphoproteomics post-translational modification (PTM) pilot on PBMCs.

3.1.4.3.1 *The recipient shall evaluate samples and methods for PTM phosphoproteomics through a small pilot to determine whether enrichment of phosphoproteins will be performed using either Titanium dioxide or antibody-based methods followed by LC-MS-based phosphoprotein analysis.*

3.1.4.3.2 *Assuming PBMC samples and methods are suitable for phosphoproteomics analysis, the recipient shall analyze PBMCs from an existing influenza H1N1 challenge repository for a pilot analysis (n= up to 60 samples).*

3.1.4.3.3 *(12 Month) Milestone: Post-translational modification pilot on existing influenza H1N1 challenge repository samples completed.*

3.1.4.4 (12 Month) Deliverable: Prospective molecular, cellular, and microbiologic datasets generated.

3.1.5 DU – Compile, curate, and annotate generated data from prospective collections.

THE TASK (3.1.5) SHALL NOT BE EXERCISED AND TASKING FUNDS RELEASED UNTIL IRB DOCUMENTATION AND PROPER IRB APPROVAL HAS BEEN OBTAINED.

The recipient shall collect, annotate, and aggregate data from the Duke challenge study to inform the selection of assays with promising biomarker data to be used for model development and biomarker discovery in Technical Area (TA) II.

3.1.5.1 The recipient shall compile and curate data from the Duke influenza challenge.

3.1.5.1.1 *The data shall include MSD cytokines (plasma and nasal lavage), RNA sequencing data from nasal curettage, single-cell (PBMC) RNA sequencing data, and post-translational modification data.*

3.1.5.1.2 *(12 Month) Milestone: Fully curated and annotated clinical, physiologic, molecular, and laboratory data from Tasks 3.1.3 and 3.1.4.*

3.1.5.2 The recipient shall aggregate, normalize, and co-register Duke challenge data.

3.1.5.2.1 *The recipient shall perform standard QC procedures and produce QC metrics and reports to complement curated data.*

3.1.5.2.2 *(12 Month) Milestone: Standard QC procedures completed and QC reports developed for the study and molecular datasets.*

3.1.5.3 The recipient shall enter Duke challenge data into the database and perform standard checks to ensure data integrity.

3.1.5.3.1 *(12 Month) Milestone: Data entry completed and data integrity checks verified.*

3.1.5.3.2 *(12 Month) Deliverable: Federated database locked and secured access available for analyses.*

3.1.6 DU –Analyze pilot data to for down-selection of assays and time points.

The recipient shall analyze each individual component of the data generated in Task 3.1.4 using standard univariate and multivariate statistical methods for molecular data to assess signal strength with respect to the outcome variable (composite of shedding and symptoms data, and supplemental physiologic and auditory data). The recipient shall evaluate the predictive power of each of the assays considered and shall quantify the importance of each time point either as predictor or (subject-level) normalizer.

3.1.6.1 The recipient shall perform statistical analysis of the molecular datasets generated in Task 3.1.4.

3.1.6.1.1 The recipient shall analyze each dataset in isolation with the goal of quantifying their individual predictive capabilities.

3.1.6.1.2 The recipient shall assess signal strength at the analyte level and as a combination of analytes (multivariate model).

3.1.6.1.3 The recipient shall measure dataset quality in terms of univariate/multivariate signal strength, dynamic range of the signal, and proportion of analytes with predictive ability (for robustness).

3.1.6.1.4 (5 Month) – Milestone: Molecular pilot datasets evaluated for signal strength, dynamic range, and analyte predictability.

3.1.6.2 The recipient shall perform statistical analysis of the sample collection time points of the challenge study.

3.1.6.2.1 The recipient shall assess the importance of each of the challenge time points considered in terms of their ability to serve as inputs to predictive models and their value as a stabilizing signal to be used as a subject-level normalizing factor.

3.1.6.2.2 (5 Month) Milestone: Each time point evaluated for value in predictive models and stabilizing or normalizing factor.

3.1.6.3 The recipient shall make recommendations for a larger study.

3.1.6.3.1 The recipient shall prepare a report compiling their findings and shall summarize assay and data point recommendations for a larger study.

3.1.6.3.2 (5 Month) Deliverable: Report of pilot analyses and recommendation for larger analyses.

Technical Area II (TA II)

3.1.7 DU – Registration, normalization and preprocessing of multi-study multimodal data.

3.1.7.1 The recipient shall perform subject-specific time-series normalization methods to individual modalities (e.g., transcriptomics, cytokines, etc.).

3.1.7.1.1 (14 Month) Milestone: Normalized subject-specific time series of individual modality datasets.

3.1.7.2 The recipient shall complete quality control, outlier identification, distribution checks, missing data identification, multimodal data registration (cross-modality time alignment and normalization) as data becomes available.

3.1.7.3 The recipient shall conduct normalization, preprocessing and registration of influenza data from Duke and additional multimodal datasets collected and generated by other performers.

3.1.7.3.1 This includes, QC, filtering (outliers and excessive missing data), missing data imputation, subject-by-subject time series normalization, batch effects assessment and *correction of cross-study effects via robust normalization.*

3.1.7.4 (14 Month) Milestone: QC, outlier identification, distribution checks, missing data, and multimodal data registration performed.

3.1.7.5 (14 Month) Deliverable: Normalized and preprocessed datasets generated and available for analyses.

3.1.8 DU, UM – Construct multimodal and physiological predictors of contagiousness.

3.1.8.1 The recipient shall design baseline high-performance predictor models over the multimodal time series organized and preprocessed in Task 3.1.7.

3.1.8.1.1 These models use shared sparsity to perform simultaneous variable selection and contagion prediction.

3.1.8.1.2 The recipient shall extend the personalized reference-based sparse prediction framework to accommodate multiple baseline samples, multimodality data, and quickest detection, missing data, and multiple batch effects.

3.1.8.2 The recipient shall combine base predictors together, including the predictors developed by Hero and Henao, to achieve enhanced performance.

3.1.8.2.1 The recipient shall use simulated ground truth and experimental data to determine prediction accuracy using standard cross-validation training-test holdout techniques.

3.1.8.2.2 The recipient shall train the predictor on shedding and symptom data.

3.1.8.2.3 The recipient shall demonstrate that a predictor trained on complex multimodality time series data can achieve at least 75% prediction accuracy over simulated and retrospectively collected data.

3.1.8.2.4 (16 Month) Milestone: Combined based predictors to enhance performance.

3.1.8.2.5 (16 Month) Milestone: Prediction accuracy determined from simulated ground truth and experimental data.

3.1.8.2.6 (16 Month) Deliverable: A multimodality signature realized that achieves 75% accuracy in predicting shedding/symptomatic infection within 24 hours of infection.

3.1.9 DU, UM – Apply multimodality predictor to prospective data to achieved desired prediction accuracy.

3.1.9.1 The recipient shall apply the predictor designed in Task 3.1.8 to achieve the Prometheus program milestones on both the retrospective and prospective data.

3.1.9.1.1 The recipient shall use measures of area under the curve (AUC) that are cross-validated on the challenge study datasets to demonstrate 75% and 95% accuracy in predicting shedding/symptomatic infection within 24 hours of infection.

3.1.9.1.2 (18 Month) Milestone: Use of AUC measures in a multimodality predictive model.

3.1.9.1.3 The recipient shall demonstrate a predictive model that predicts shedding/symptomatic infection within 24 hours of infection for at least 75% of the subjects and for at least 95% of the subjects.

3.1.9.1.4 (18 Month) Milestone: Integration of multimodal predictive models to achieve program goals.

3.1.9.1.5 (18 Month) Deliverable: A multimodality predicting signature realized with 95% accuracy in predicting shedding/symptomatic infection within 24 hours of infection.

3.1.10 DU – Project Management

***3.1.10.1** Deliverable: The recipient shall provide a final summary report and an IRB closeout/final report.*

3.2 OPTION PERIOD (Phase II)

3.2.1 DU – Expanded molecular assays to additional time points.

***3.2.1.1** The recipient shall perform cytokine assays on additionally curated and quality control samples from the DARPA Prometheus consortium (up to n = 600).*

***3.2.1.2** The recipient shall extend RNA extraction protocol to additional samples including paxgenes, nasal curettage, and single cells collected by the consortium (up to n = 700)*

3.3 OPTION PERIOD (Phase III)

3.3.1 DU—Enroll an additional 10 volunteer subjects

***3.3.1.1** The recipient shall enroll and challenge an additional 10 volunteer subjects for a cohort of 50 subjects total.*

3.4 Milestones

Phase I. Base Period

3.4.1 DU - Generate pilot metabolomics data.

3.4.1.1.1 (3 Month) Milestone: Curated list of existing samples from Duke cohorts.

3.4.1.2.1 (3 Month) Milestone: Curated list of existing samples from possible consortium cohorts.

3.4.1.3.1 (3 Month) Milestone: Pilot data generated on 3 or more metabolomics platforms (may include but not limited to p180, lipids, and oxylipin/oxysterols, energy metabolite panel, or purine/pyrimidine panel).

3.4.1.4.1 (3 Month) Milestone: QC, normalized, and packaged pilot dataset for analyses.

3.4.2 DU – Design and build federated database structure of Duke data.

3.4.2.1.1 (4 Month) Milestone: Developed SOPs for data handling, database design, structure, and content of dictionaries.

3.4.2.2.2 (4 Month) Milestone: Database implementation and testing complete.

3.4.2.3.1 (4 Month) Milestone: Sharing of user protocols developed and launched.

3.4.2.4.1 (4 Month) Milestone: Database populated and tested with pilot dataset from Task 3.1.1

3.4.3 DU, ICL, UNC – Execute Influenza challenge study (number of subjects = 40)

3.4.3.2.1 (8 Month) Milestone: Regulatory approvals obtained from U Ethics, Duke IRB, and HRPO.

3.4.3.3.1 (8 Month) Milestone: Up to 250 subjects screened for pH1N1 antibodies by HA.

3.4.3.4.5 (8 Month) Milestone: Up to 40 subjects challenged with pH1N1, and time-series bio-specimens, clinical, physiologic, audio, and laboratory data collected.

3.4.4 DU – Generate molecular data from prospective sample collections at meaningful time points.

3.4.4.1.4 (12 Month) Milestone: RNA extractions prepared from peripheral blood, PBMCs, and nasal curettage samples for RNASeq~~.,~~ *and complete RNA sequencing from single immune cell preparations from peripheral blood.*

3.4.4.2.2 (12 Month) Milestone: 30-panel MSD assays completed on peripheral blood and nasal lavage samples.

3.4.4.3.3 (12 Month) Milestone: Down-selected metabolomics assay completed on consortium samples.

3.4.4.4.2 Milestone: Post-translational modification pilot on existing influenza H1N1 challenge repository samples completed.

3.4.5 DU – Compile, curate, and annotate generated data from prospective collections.

3.4.5.1.2 (12 Month) Milestone: Fully curated and annotated clinical, physiologic, molecular, and laboratory data from Tasks 3.1.3 and 3.1.4.

3.4.5.2.2 (12 Month) Milestone: Standard QC procedures completed and QC reports developed for the study and molecular datasets.

3.4.5.3.1 (12 Month) Milestone: Data entry completed and data integrity checks verified.

3.4.6 DU –Analyze pilot data to for down-selection of assays and time points.

3.4.6.1.4 (5 Month) Milestone: Molecular pilot datasets evaluated for signal strength, dynamic range, and analyte predictability.

3.4.6.2.2 (5 Month) Milestone: Each time point evaluated for value in predictive models and stabilizing or normalizing factor.

3.4.7 DU – Registration, normalization and preprocessing of multi-study multimodal data.

3.4.7.1.1 (14 Month) Milestone: Normalized subject-specific time series of individual modality datasets.

3.4.7.4 (14 Month) Milestone: QC, outlier identification, distribution checks, missing data, and multimodal data registration performed.

3.4.8 DU, UM – Construct multimodal and physiological predictors of contagiousness.

3.4.8.2.4 (16 Month) Milestone: Combined based predictors to enhance performance.

3.4.8.2.5 (16 Month) Milestone: Prediction accuracy determined from simulated ground truth and experimental data.

3.4.9 DU, UM – Apply multimodality predictor to prospective data to achieved desired prediction accuracy.

3.4.9.1.2 (18 Month) Milestone: Use of AUC measures in a multimodality predictive model.

3.4.9.1.4 (18 Month) Milestone: Integration of multimodal predictive models to achieve program goals.

4.0 Deliverables:

Phase I. Base Period

4.1 (3 Month) Deliverable: Fully curated and annotated pilot dataset generated and ready for analyses in Task 3.1.6.

4.2 (4 Month) Deliverable: Functional and access-secured federated database generated.

4.3 (8 Month) Deliverable: Study samples and associated clinical, physiologic, and laboratory data secured and housed in Duke biobank and data system.

4.4 (12 Month) Deliverable: Prospective molecular, cellular, and microbiologic datasets generated.

4.5 (12 Month) Deliverable: Federated database locked and secured access available for analyses.

4.6 (5 Month) Deliverable: Report of pilot analyses and recommendation for larger analyses.

4.7 (14 Month) Deliverable: Normalized and preprocessed datasets generated and available for analyses.

4.8 (16 Month) Deliverable: A multimodality signature realized that achieves 75% accuracy in predicting shedding/symptomatic infection within 24 hours of infection.

4.9 (18 Month) Deliverable: A multimodality predicting signature realized with 95% accuracy in predicting shedding/symptomatic infection within 24 hours of infection.

4.10 Deliverable: The recipient shall provide a final cooperative agreement summary report and an IRB closeout/final report.

5.0 PROGRAM MANAGEMENT AND REVIEW

5.1 Kick-off Meeting.

The recipient shall hold a kick off meeting within 60 days of award of this cooperative agreement. In this meeting, the recipient shall present a program management plan and financial tracking plan. Presentation materials for this meeting shall be provided.

5.2 Monthly Financial Reports.

The recipient shall provide monthly financial progress reports to SSC Pacific and DARPA. The purpose of these reports is to provide a brief project progress and inform DARPA Program Manager of any potential issues.

5.3 Quarterly Technical Reporting.

The recipient shall provide quarterly progress reports to SSC Pacific and DARPA. The purpose of these reports is to present a summary of work completed by SOW tasking and milestones met, discuss any problems encountered, update the program schedule, present the program financial status, and discuss remaining work.

5.4 Final Cooperative Agreement Review.

The recipient shall host a final cooperative agreement review. The purpose of this review is to present a summary of all work completed and milestones accomplished and to discuss any relevant future efforts similar to the cooperative agreement, which may be pursued. This report shall be provided to SSC Pacific and DARPA. A final cooperative agreement summary report shall be provided no later than 60 days after the end of the program.

5.5 Government Management/Review.
The Government will actively monitor, review and approve the recipient's performance to ensure all the performers are in sync and matched with the Government's requirements. The Government will ensure that each of the performers share experimental data across the program. The Government will further ensure that the performers develop techniques and capabilities that are compatible and integrate with each other. The recipient shall collaborate and cooperate with other performers in the program under the direction of the Government. At the Government Principal Investigator (PI) meeting, all performers shall demonstrate their technical capabilities and engage or challenge each other in a cooperative and challenging environment. Along these lines, the Government will ensure that the performers each share technical information with each other to enable the testing/challenging of each other's capabilities. The Government will further oversee the program and will review, approve, and participate in the demonstrations, as well as determine which performers will advance to the Option Phase(s).

5.6 Data sharing.
Data sharing is anticipated in this effort. The recipient shall support data sharing with other performers including sharing of raw and processed experimental data, processing methods used, algorithms used to process the data, artifact information, research reports, and software including source code and executables. Data sharing will be conducted in a manner that provides enough description for a third party to look at and conduct appropriate analysis and interpretation of the data.

6.0 Travel
Travel is anticipated for meetings between collaborators and meetings with DARPA.

7.0 Place Of Performance
Duke University
Imperial College, London
University of Michigan
University of North Carolina at Chapel Hill

8.0 Government Technical Representatives
Technical POC: Patrick Sims, pcsims@spawar.navy.mil, 619 553 0828
Admin POC: Amy Nehrich, anehrich@spawar.navy.mil, 619 553 5595